Flash Guard: An AI-Assisted Embedded Flash Flood Speed Detection and Early Warning System

Project Proposal - EC 6020

Group Members:

DESHAPRIYA D.M.K.N. (2022/E/027)

PREMARATHNA H.M.K.K. (2022/E/050)

GUNARATHNA J.M.C.A. (2022/E/059)

KAVISHYA M.G.C. (2022/E/086)

HEMATHILAKA R.H.C. (2022/E/114)

Introduction

Flash floods are one of the most sudden and destructive natural disasters experienced in Sri Lanka, particularly during monsoon seasons. Unlike conventional floods, which develop gradually over time, flash floods occur rapidly due to intense rainfall, sudden dam releases, or drainage blockages. This sudden onset leaves very little time for evacuation or preventive action, leading to severe damage to human lives, infrastructure, transportation networks, and agricultural lands.

Most existing flood monitoring systems focus primarily on water level measurement. While water level is an important parameter, it does not always accurately represent the severity of a flash flood. In many real-world situations, dangerous conditions arise due to a **rapid increase in water flow speed**, even before water reaches a critical height. High flow velocity significantly increases the kinetic energy of moving water, resulting in strong erosive forces that can damage bridges, roads, buildings, and other structures, while posing serious risks to human safety.

Furthermore, conventional threshold-based flood warning systems often suffer from **delayed detection and high false alarm rates**, as they are unable to adapt to changing environmental conditions. This reduces the effectiveness of early warning mechanisms and may lead to a lack of public trust in warning alerts. As climate change continues to increase the frequency and intensity of extreme weather events, the limitations of traditional flood monitoring approaches become more evident.

Therefore, there is a clear need for a **real-time, intelligent, and low-cost embedded solution** that focuses on detecting sudden changes in water flow behavior rather than relying solely on static water level thresholds. This project proposes **FlashGuard**, an AI-assisted embedded system that continuously monitors water flow speed, analyzes dynamic flow patterns using intelligent decision logic, and provides early warnings through visual, audible, and digital interfaces. The proposed system aims to enhance early flood detection capability, reduce disaster-related risks, and support effective decision-making in flood-prone regions of Sri Lanka.

Design Overview

- **System Architecture :**

The proposed system is designed as a layered embedded solution integrating hardware sensing, intelligent processing, and user interaction. The architecture consists of three main layers:

1. Sensing Layer

This layer is responsible for collecting real-world environmental data. A **Hall-effect water flow sensor** is placed in the water channel. As water flows, the sensor generates electrical pulses proportional to the speed of water flow.

2. Processing and Intelligence Layer

A **standalone ATmega328P microcontroller**, programmed in embedded C/C++, processes the sensor pulses. Using interrupt-based pulse counting and timer mechanisms, the microcontroller calculates real-time flow speed and extracts features such as rate of change and short-term averages. An **AI-assisted decision logic** then analyzes these features to classify flood conditions.

3. Alert and User Interface Layer

Based on the system decision, alerts are generated using:

- LEDs for visual indication (Normal, Warning, Flash Flood)
- A buzzer for audible alerts
- A serial-based user interface (desktop or web application) for real-time monitoring

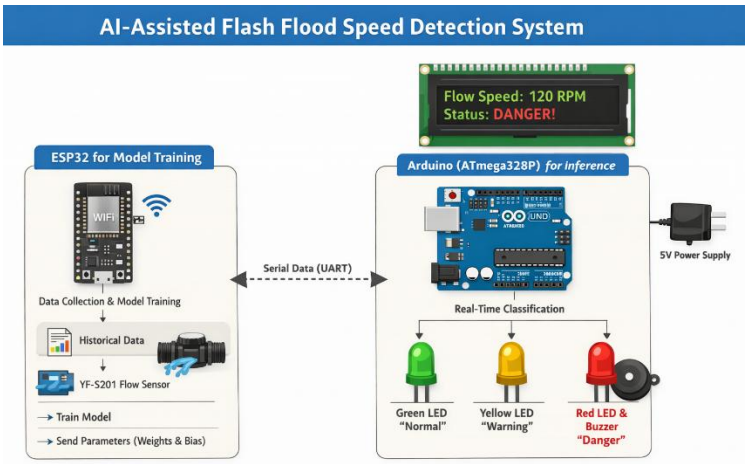

Figure 01: High-level hardware–software interaction of the Flash Guard system

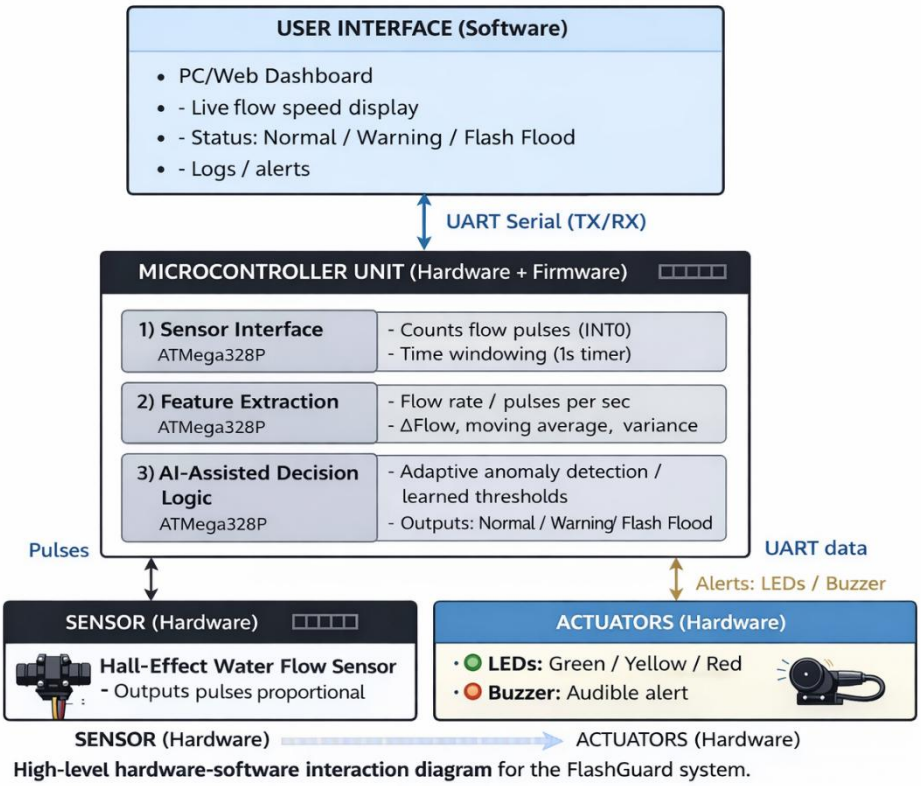

Figure 02: High-level hardware–software interaction of the Flash Guard system.

- **Communication Protocols :**

The system uses optimized communication protocols to reduce wiring complexity and ensure reliable data transfer.

I2C (Inter-Integrated Circuit)

- Used for communication with optional display modules or additional sensors
- Requires only two wires (SDA, SCL)
- Supports multiple devices on the same bus
- Minimizes hardware complexity

UART (Universal Asynchronous Receiver-Transmitter)

- Used for communication between the microcontroller and the user interface
- Enables real-time data transmission to a desktop or web application
- Simple, reliable, and widely supported

GPIO with Interrupts

- Used for pulse input from the water flow sensor
- Ensures accurate real-time detection of flow changes
- Essential for time-critical embedded applications

The use of these protocols aligns with the guideline requirement to optimize connections and minimize wiring.

- ## Circuit Design :

The circuit for the **AI-Assisted Flash Flood Speed Detection System** integrates several components to achieve real-time flood detection using the **ATmega328P microcontroller** and **ESP32** for model training. The circuit also includes a **LCD display** to visualize the current flow conditions and status indicators, including **LEDs** for different flood stages.

Key Components:

1. **ESP32**: Responsible for data collection and **training the AI model**. It communicates with the **ATmega328P** to transfer trained model parameters.
2. **ATmega328P**: The core microcontroller that performs **real-time flood classification**. It receives the model parameters from the **ESP32** and uses them to classify the water flow data from the **YF-S201 flow sensor**.
3. **LCD Display (16x2)**: Displays the current **flow speed** and **classification** (Normal, Warning, Danger) in real-time.
4. **YF-S201 Water Flow Sensor**: Measures the **flow speed** of water, which is used as an input to the classification model.
5. **LED Indicators**: **Green, Yellow, and Red LEDs** are used to indicate the flood status based on the classification result:
    - **Green LED**: Normal flow
    - **Yellow LED**: Warning condition
    - **Red LED**: Flash flood condition
6. **Buzzer**: Provides an **audible warning** when a **flash flood** condition is detected.
7. **Capacitors**: Used for **stabilizing the power supply** and ensuring proper functionality of the microcontroller and components.
8. **Transistor (BC547)**: Used to drive the **buzzer** based on the flow status signal.

Diagram Description:

- The **ESP32** acts as the central unit for **training the machine learning model** and **communicating with the ATmega328P**.
- **ATmega328P** is used to **process real-time data** from the flow sensor and classify it according to the trained model.

- The **LCD display** shows **current flow conditions** and **classification results**, providing a user-friendly interface for real-time monitoring.
- **LEDs** provide a clear visual representation of the flood condition, while the **buzzer** alerts users to a flash flood condition.

This circuit design efficiently integrates the **AI model**, **real-time data processing**, and **user feedback mechanisms** (via the display and LEDs) for flood detection.

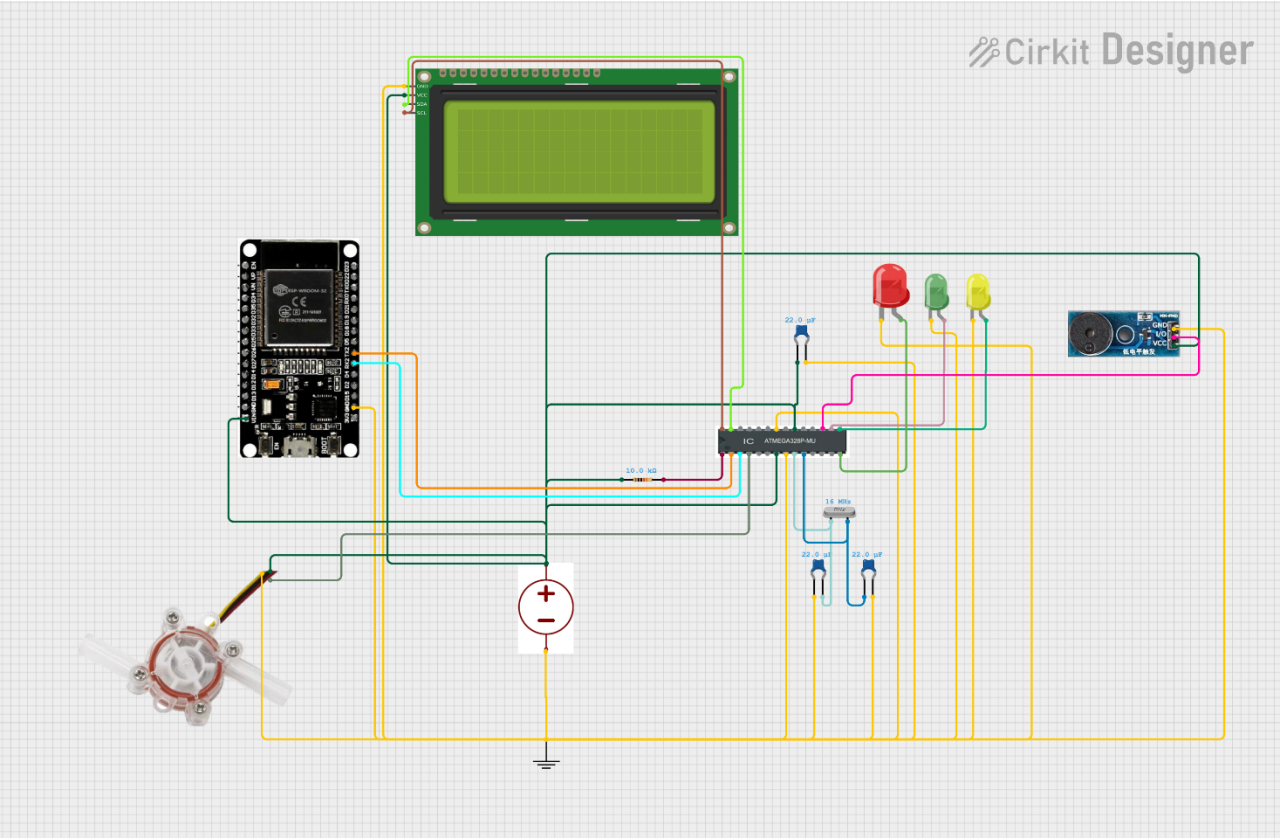

Figure 03: Circuit Diagram

Technologies to Be Used

Hardware Technologies

- ATmega328P Microcontroller
- ESP32
- Hall-effect Water Flow Sensor (YF-S201 or equivalent)
- LEDs (Red, Yellow, Green)
- Active Buzzer
- NPN Transistor (2N2222 / BC547)
- Passive components (resistors, power supply)
- Breadboard and jumper wires

Software Technologies

- Embedded C/C++ for microcontroller programming
- Arduino IDE / AVR-GCC (compiler only)
- Python for AI model training and data analysis

Uniqueness and Innovation

Innovative Features :

- Focus on **flow speed** rather than only water level
- AI-assisted adaptive decision making
- Fully standalone operation without internet dependency
- Low-cost and scalable design

Comparison with Existing Solutions :

Existing Systems	Proposed System
Level-based detection	Speed-based detection
Fixed thresholds	AI-assisted adaptive logic
High cost	Low cost
Late warnings	Early flash flood detection

This project introduces an innovative **AI-Assisted Flash Flood Speed Detection System** that leverages the combined processing power of two microcontrollers: the **ATmega328P** for real-time flood detection and the **ESP32** for model training. The uniqueness of this system lies in its **adaptive flood detection approach** and the **use of machine learning** for more accurate predictions.

Unlike conventional flood monitoring systems, which often rely on simple threshold-based detection, this project utilizes a **trained machine learning model** to classify water flow conditions based on data from the **YF-S201 flow sensor**. The **ESP32** handles the **data collection and model training** by processing **historical data** or real-time sensor input, which is then used to **train a machine learning model**. The trained model parameters—**weights and bias**—are transferred to the **ATmega328P** for **real-time inference**.

The system classifies water flow into three categories: **Normal**, **Warning**, and **Danger**. Depending on the classification, the system activates corresponding **visual indicators (LEDs)** and an **audible alert (buzzer)** to provide immediate feedback on flood risk levels. The **use of an LCD display** further enhances the system's user interface by showing real-time **flow speed** and the **flood status**.

Budget

Component	Average Cost (LKR)	Quantity	Total Average Cost (LKR)
ATmega328P Microcontroller	850	1	850
ESP32	1,850	1	1,850
YF-S201 Water Flow Sensor	475	1	475
LCD Display (16x2)	850	1	850
LEDs (Green, Yellow, Red)	40	3	120
330Ω Resistors	15	3	45
NPN Transistor (BC547/2N2222)	45	1	45
Active Buzzer	175	1	175
Capacitors (100nF, 22pF)	20	3	60
Resistors (1kΩ, 10kΩ)	15	2	30
5V Power Supply (Adapter)	350	1	350

Project Timeline

Week	Phase
Week 1	Project Initiation
Week 2	Requirement Analysis
Week 3	Proposal Finalization
Week 4	Hardware Planning
Week 5	Hardware Implementation
Week 6	Embedded Software – I
Week 7	Embedded Software – II
Week 8	MID Evaluation
Week 9	AI Logic Integration
Week 10	User Interface Development
Week 11	System Integration
Week 12	Testing & Refinement
Week 13	END Evaluation
Week 14	Documentation
Week 15	Buffer & Improvements

****